

AAYUSH PANDEY

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Education

VIT Bhopal University

Aug 2023 – Aug 2027

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.24

Madhya Pradesh

Experience

DRDO – Summer Trainee | [Github](#) | [Live Demo](#) | Apr 2026 – Jun 2026

(Python, TensorFlow, TensorFlow Lite, MediaPipe, face-api.js, React, Node.js, Express.js, PostgreSQL)

- Developed a real-time ASL-to-text communication platform with ~99% gesture recognition accuracy across 26 hand signs and <100 ms latency, integrating MediaPipe, TensorFlow, face-api.js, and React for gesture and face recognition.
- Engineered a secure backend with 6 RESTful APIs, JWT-based authentication, an admin dashboard, and a PostgreSQL database supporting 127 facial descriptors and user/admin records for real-time identification.

DRDO – Winter Trainee | [Github](#) | Nov 2025 – Jan 2026

(Python, React, Node.js, Express.js, PostgreSQL, Prisma ORM, Ollama (Llama 3), REST APIs)

- Engineered a full-stack OSINT investigation platform that automated entity extraction from documents and metadata, reducing manual investigative effort by 75%, while integrating Ollama (Llama 3) for iterative keyword extraction and AI-generated investigation reports.
- Designed a normalized PostgreSQL schema for 6+ core entities and developed 5+ OSINT automation scripts to enrich intelligence from public sources, enabling efficient relationship mapping and streamlined investigations.

Projects

MindConnect | [Github](#) | [Live Demo](#)

(TypeScript, React, Node.js, Express.js, Prisma ORM, PostgreSQL (Neon), JWT Authentication, Vercel)

- Architected and deployed a full-stack **mental health platform** with **10+ RESTful API endpoints** covering authentication, appointments, admin, and doctor management, implementing **JWT-based authentication** and **role-based access control** to secure all protected routes.
- Designed a normalized **PostgreSQL** database using **Prisma ORM** to improve data consistency across **5 core modules**, and deployed the application on **Vercel** with **Neon PostgreSQL** using a **TypeScript monorepo** with shared types across the client and server.

MediScan | [Github](#) |

(TypeScript, React, Node.js, Express.js, Prisma ORM, PostgreSQL (Neon), Vercel)

- Developed a desktop medical imaging application using Electron.js and Python to analyze DICOM CT/MRI scans, implementing a 7-stage image processing pipeline for automated muscle and fat tissue segmentation with interactive mask refinement.
- Engineered classical computer vision algorithms including intensity thresholding, Otsu segmentation, non-local means denoising, morphological operations, connected-component analysis, and pixel-spacing-based area calculation, with results stored in SQLite.

Technical Skills

Programming Languages: Python, Java, JavaScript, HTML, CSS, SQL

Frameworks & Runtime: React, Node.js, Express.js, Tailwind CSS, Electron.js, RESTful APIs

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Oracle SQL

ML & Data: TensorFlow, TensorFlow Lite, Keras, OpenCV, MediaPipe, face-api.js, NumPy, scikit-image, pydicom

Developer Tools: VS Code, Vite, Linux (Ubuntu, Arch, Kali), Bash

Cloud & DevOps: AWS (EC2, S3, IAM), Docker, Vercel, Git, GitHub, Render

Personal details

Interests: Artificial Intelligence, Machine Learning, Web Development, Cloud Architectures, Cybersecurity, Philosophy.

Hobbies: Exploring Open-Source Technologies, Reading, Mixed Martial Arts.

Extracurricular: Active Linux user (Ubuntu, Arch, Kali).

Certificates

Marketing Analysis(Elite + Gold) | NPTEL | Apr 2026

Google IT Support Professional Certificate | Google via Coursera | Jan 2026

Introduction to Machine Learning | NPTEL | Apr 2025